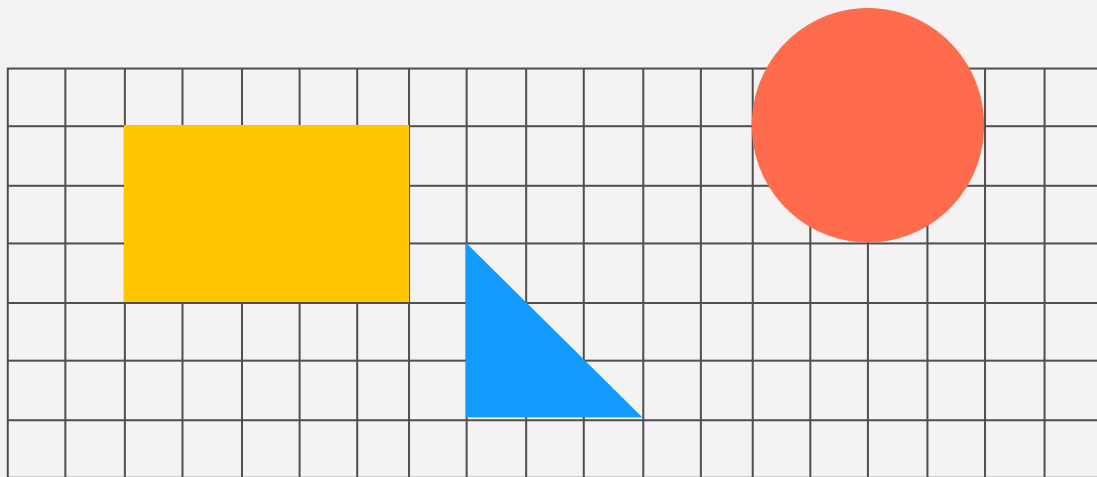


Fairness Constrained Logistic Regression for Automated Resume Screening

A convex
Optimization Approach

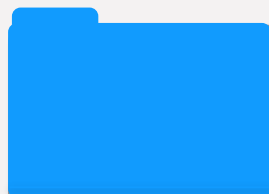


Introduction



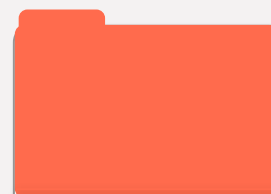
- Problem Identification

Problem Modeling



- Logistic Regression Baseline Setup
- Proof of Convexity
- Fairness Constraint
- Why It breaks Convexity
- Lagrangian Relaxation & KKT

Implementation & Results



- Dataset & Bias Visualization
- Accuracy-Fairness Trade-Off
- Results
- Interpretability

Conclusion



- Limitations
- Conclusion
- Future Work

Problem Identification:

What is Automated Resume Screening?

Research Question

Can we **formulate** resume screening as a **constrained convex optimization problem** that is both **accurate** and **provably fair**?



What is Automated Resume Screening

- Companies use ML models to **automatically filter** thousands of job applications
- Models are trained on **historical hiring decisions**
- Your resume may be rejected before a **human** ever sees it

The Fairness Problem

- Historical hiring data reflects existing demographics biases
- A model optimized purely for accuracy will reproduce and perpetuate those disparities
- Equally qualified candidates from different groups get treated unequally

Problem Modeling: Setup & Baseline

Logistic Regression Baseline

Decision Variable:

$\beta \in R^n$ (regression coefficient vector)

Constants & Data:

- m training samples: $\{(x_i, y_i, g_i)\}_{i=1}^m$
- $x_i \in R^n$ - feature vector:
experience, test score,
education score, projects
- $y_i \in \{0,1\}$ - true shortlisting label
- $g_i \in \{0,1\}$ - sensitive group attribute
(demographic group)
- $\lambda > 0$ - L2 regularization constant
- $\Upsilon \geq 0$ - fairness penalty constant

Model - Shortlisting Probability:

$$P(y_i = 1|x_i) = \sigma(\beta^T x_i), \text{ where } \sigma(z) = 1/(1 + e^{-z})$$

Baseline Objective - Regularized Log-Loss:

$$\min_{\beta} L(\beta) + \lambda \|\beta\|_2^2$$

Where the log-loss is:

$$L(\beta) = \left(\frac{1}{m}\right) \sum_{i=1}^m m[-y_i \log \sigma(\beta^T x_i) - (1 - y_i) \log(1 - \sigma(\beta^T x_i))]$$

No constraints – purely accuracy-driven.
This is the model we need to improve upon for fairness.

Convexity Proof of the Baseline

Proof of Convexity – Baseline Objective

Log-loss $L(\beta)$ is convex:

Key identity: $\sigma'(z) = \sigma(z)(1 - \sigma(z))$

Let $p_i = \sigma(\beta^T x_i)$.

Compute the Hessian of a single sample loss $l_i(\beta)$:

$$\nabla_{\beta}^2 l_i(\beta) = p_i(1 - p_i) x_i x_i^T$$

For any $v \in \mathbb{R}^n$:

$$v^T \nabla_{\beta}^2 l_i(\beta) v = p_i(1 - p_i) (x_i^T v)^2 \geq 0$$

Since $p_i \in (0,1) \rightarrow p_i(1 - p_i) > 0$, and $(x_i^T v)^2 \geq 0$

$\therefore \nabla^2 l_i \geq 0 \rightarrow l_i$ is convex

$$L(\beta) = \left(\frac{1}{m}\right) \sum_i l_i(\beta) \text{ is convex}$$

Non-negative sum of convex functions. $\checkmark$

Claim:

$$F(\beta) = L(\beta) + \lambda \|\beta\|_2^2$$

is strictly convex in β

Regularizer $\lambda \|\beta\|_2^2$ is strictly convex:

$$\nabla_{\beta}^2 (\lambda \|\beta\|_2^2) = 2\lambda I$$

For any $v \neq 0$:

$$v^T (2\lambda I) v = 2\lambda \|v\|^2 > 0 \text{ (since } \lambda > 0 \text{)}$$

$\therefore \lambda \|\beta\|_2^2$ is strictly convex. $\checkmark$

Combined objective:

$$F(\beta) = L(\beta) + \lambda \|\beta\|_2^2 = \text{convex} + \text{strictly convex}$$

$\therefore F(\beta)$ is strictly convex

$\rightarrow$ unique global minimum exists and is guaranteed. $\checkmark$

Solved using: L-BFGS-B

(gradient based, exploits smoothness of F)

Fairness Constraint: Why it Breaks Convexity

The Equal Opportunity Constraint

● ● ● Fairness Metric TPR per group:

For group $k \in \{0,1\}$, define $S_k = \{i: g_i = k, y_i = 1\}$
(truly qualified candidates in group k):

$$TPR_k(\beta) = \left(\frac{1}{|S_k|} \right) \sum_{i \in S_k} \sigma(\beta^T x_i)$$

Hard Fairness Constraint (Equal Opportunity):

$$|TPR_0(\beta) - TPR_1(\beta)| \leq \varepsilon$$

Full Constrained Problem:

$$\begin{aligned} & \min_{\beta} L(\beta) + \lambda \|\beta\|_2^2 \\ & \text{subject to: } |TPR_0(\beta) - TPR_1(\beta)| \leq \varepsilon \end{aligned}$$

● ● ● Is the constraint convex?

Define $h(\beta) = TPR_0(\beta) - TPR_1(\beta)$

= linear combination of $\sigma(\beta^T x_i)$

Compute the Hessian of a single $\sigma(\beta^T x_i)$ term:

$$\nabla_{\beta}^2 \sigma(\beta^T x_i) = p_i(1-p_i)(1-2p_i) x_i x_i^T$$

The scalar factor $p_i(1-p_i)(1-2p_i)$ changes sign depending on p_i :

- When $p_i < 0.5 \rightarrow \text{factor} > 0$ (locally convex)
 - When $p_i > 0.5 \rightarrow \text{factor} < 0$ (locally concave)
- $\therefore \sigma(\beta^T x_i)$ is neither convex nor concave

$h(\beta)$ is non convex

$\rightarrow$ the constraint $|h(\beta)| \leq \varepsilon$ is non convex

Fairness Constraint: Why it Breaks Convexity

The Equal Opportunity Constraint

Fairness Metric TPR per group:

For group $k \in \{0,1\}$, define $S_k = \{i: g_i = k, y_i = 1\}$

(truly qualified candidates in group k)

$$TPR_k(\beta) = \left(\frac{\sum_{i \in S_k} \sigma(\beta^T x_i)}{\sum_{i \in S_k} 1} \right)$$

Hard Fairness Constraint

$$|TPR_0(\beta) - TPR_1(\beta)| \leq \varepsilon$$

Full Constrained Problem

$$\min_{\beta} L(\beta) + \lambda \|\beta\|_2^2$$

$$\text{subject to: } |TPR_0(\beta) - TPR_1(\beta)| \leq \varepsilon$$

Consequence:

The full **constrained problem is non-convex.**

We **cannot** directly **guarantee a global optimum.**

We **need a principled relaxation.**

Is the constraint convex?

Define $h(\beta) = TPR_0(\beta) - TPR_1(\beta)$

= linear combination of $\sigma(\beta^T x_i)$

single $\sigma(\beta^T x_i)$ term:

$$\sigma(\beta^T x_i) = \frac{1}{1 + \exp(-\beta^T x_i)}$$

$(1 - 2p_i) x_i x_i^T$ changes sign

(locally convex)

• When $p_i \rightarrow \text{factor} < 0$ (locally concave)

$\therefore \sigma(\beta^T x_i)$ is neither convex nor concave

$h(\beta)$ is non convex

$\rightarrow$ the constraint $|h(\beta)| \leq \varepsilon$ is non convex

Lagrangian Relaxation & Penalized Objective

Our Optimization Approach

Standard Lagrangian:

$$\mathcal{L}(\beta, \mu)$$

$$= L(\beta) + \lambda \|\beta\|_2^2 + \mu \cdot (|TPR_0(\beta) - TPR_1(\beta)| - \varepsilon)$$

Where $\mu \geq 0$ is the Lagrange multiplier.

We use a smooth squared penalty

(differentiable, amenable to gradient methods):

$$\min_{\beta} \mathcal{L}_Y(\beta)$$

$$= L(\beta) + \lambda \|\beta\|_2^2 + Y \cdot (TPR_0(\beta) - TPR_1(\beta))^2$$

Where $Y \geq 0$ plays the role of the Lagrange multiplier μ .

KKT Conditions

1. Stationarity:

$$\begin{aligned} \nabla_{\beta} L(\beta^*) + 2\lambda\beta^* + 2\mu^* \cdot (TPR_0(\beta^*) - TPR_1(\beta^*)) \\ \cdot \nabla_{\beta} (TPR_0 - TPR_1) = 0 \end{aligned}$$

2. Primal feasibility:

$$|TPR_0(\beta^*) - TPR_1(\beta^*)| \leq \varepsilon$$

3. Dual feasibility:

$$\mu^* \geq 0$$

4. Complementary slackness:

$$\mu^* \cdot (|TPR_0(\beta^*) - TPR_1(\beta^*)| - \varepsilon) = 0$$

These are **necessary conditions** for optimality. Since the constraint is non-convex, they are not sufficient for global optimality, but our penalized approach converges to a **KKT point**.

Dataset & Bias Visualization

Dataset & The Bias Problem

Synthetic Dataset:

- 2,500 candidates
- 5 features
- 2 demographic groups

Data Features

Feature	Description
Experience	Years of work experience
Test Score	Standardized aptitude score
Education Score	Academic performance
Certifications	Number of professional certifications
Project	Number of completed projects

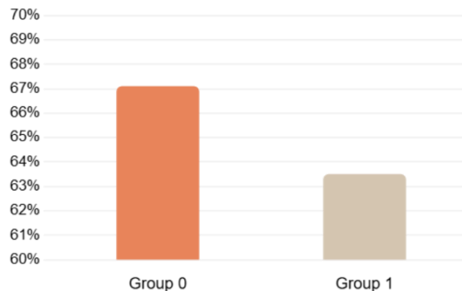
Data Features

Intentional bias injected into label generation:
Group 0 candidates are shortlisted at a higher rate even when qualifications are comparable:

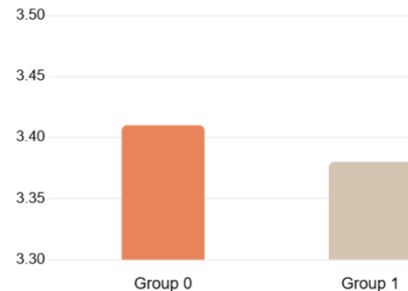
Group	Shortlisting Rate
Group 0	67.1%
Group 1	63.5%

Group 0 Group 1

Observed shortlisting rate by group



Average qualification proxy by group



Group 0 is shortlisted at a higher rate despite nearly identical qualification scores — evidence of injected historical bias.

Results: Baseline vs Fairness-Aware

Accuracy – Fairness Trade-Off

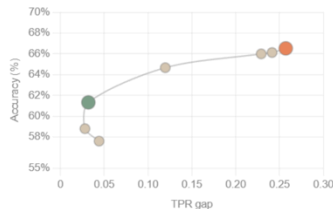
Baseline model ($\Upsilon = 0$):

- Accuracy: 66.5% | ROC-AUC: 0.755
 - TPR Group 0: 0.498 | TPR Group 1: 0.755
- TPR gap = 0.257 ❌**

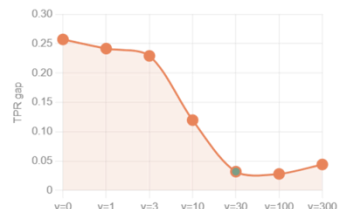
Data Features

— Baseline ($\Upsilon=0$) — Fairness-aware ($\Upsilon=30$)

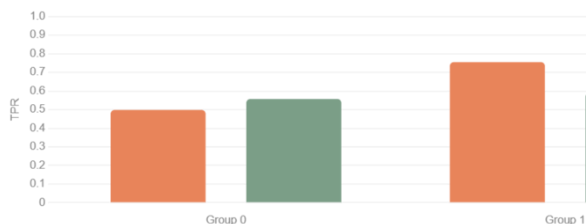
Accuracy–fairness trade-off (Pareto frontier)



TPR gap vs fairness penalty Υ



TPR by group: baseline vs fairness-aware ($\Upsilon=30$)



Sweep across Υ values:

Υ	Accuracy	ROC-AUC	TRP G0	TPR G1	TPR GAP	Fairness?
0	66.5%	0.755	0.498	0.755	0.257	❌
1	66.1%	0.757	0.494	0.735	0.242	❌
3	66.0%	0.755	0.502	0.731	0.229	❌
10	64.7%	0.724	0.544	0.664	0.120	⚠️
30	61.3%	0.672	0.557	0.589	0.032	✅
100	58.8%	0.635	0.570	0.542	0.028	✅
300	57.6%	0.621	0.574	0.530	0.044	✅

Key finding:

At $\Upsilon = 30$, the TPR gap collapses from 0.257 $\rightarrow$ 0.032 with only a modest accuracy drop of ~5 percentage points.

$\Upsilon = 100$, accuracy keeps falling but fairness does not improve further, thus diminishing returns on the Pareto frontier.

Interpretability: Coefficient Analysis

How Fairness Changes the Model

● ● ● Logistic Regression is auditable

We can directly inspect what the optimizer changed:

Feature	Baseline β	Fair β ($Y = 30$)	Interpretation
Experience	0.5678	-0.0247	Heavily down-weighted, correlated with group bias
Test Score	0.4825	0.0854	Reduced - partially biased
Certifications	0.3900	0.2154	Retained - more equitable signal
Projects	0.2959	0.0636	Reduced
Education Score	0.2162	-0.0853	Down-weighted

● ● ● What the optimizer is doing:

- The baseline heavily relies on **experience**, which is correlated with group membership in the biased dataset
- The fairness-aware model down-weights experience almost to zero and redistributes weight toward **certifications**, a more group-neutral feature
- This is exactly what the KKT stationarity condition predicts:
At optimum, the gradient of the log-loss and the gradient of the fairness penalty must balance.

Conclusion

Wrap Up

What we showed:

1. Automated resume screening encodes **historical bias** – TPR gap of 0.257 in the unconstrained baseline
2. Logistic regression is convex $\rightarrow$ unique global minimum guaranteed for the baseline
(*provided via Hessian $\succcurlyeq 0$*)
3. The equal opportunity constraint $|TPR_0(\beta^*) - TPR_1(\beta^*)| \leq \varepsilon$ is non-convex $\rightarrow$ requires Lagrangian relaxation
4. Penalized objective with $\Upsilon \cdot (TPR_{GAP})^2$ is smooth and tractable via L-BFGS-B, also sweeping Υ traces the Pareto frontier
5. At $\Upsilon = 30$: TPR gap reduced from 0.257 $\rightarrow$ 0.032 with only -5% accuracy loss
6. Coefficient analysis confirms the optimizer down-weights biased features in a principled, auditable way, consistent with KKT stationarity

Conclusion

Limitations & Future Work

Current Limitations:

1. Synthetic data with controlled bias may not reflect the complexity of real-world hiring datasets
2. The fairness constraint is non-convex, so our penalized approach only guarantees a local KKT point, not a global minimum.
3. Only TPR parity is enforced – FPR parity, calibration, and individual fairness are not addressed.
4. The binary group attribute ignores intersectionality and cannot generalize to multiple protected attributes.

Future Work:

1. Validate the model on real resume datasets to test whether the accuracy-fairness trade-off holds in practice.
2. Re-implement using CVXPY with disciplined convex programming for verified ε -tolerance guarantees.
3. Extend fairness constraints to $K > 2$ intersectional groups:
$$\max_{j,k} |TPR_j(\beta) - TPR_k(\beta)| \leq \varepsilon$$
4. Systematically compare pre-processing, in-processing, and post-processing fairness intervention strategies on the same dataset.

This is the
end of our
presentation...



Thank you